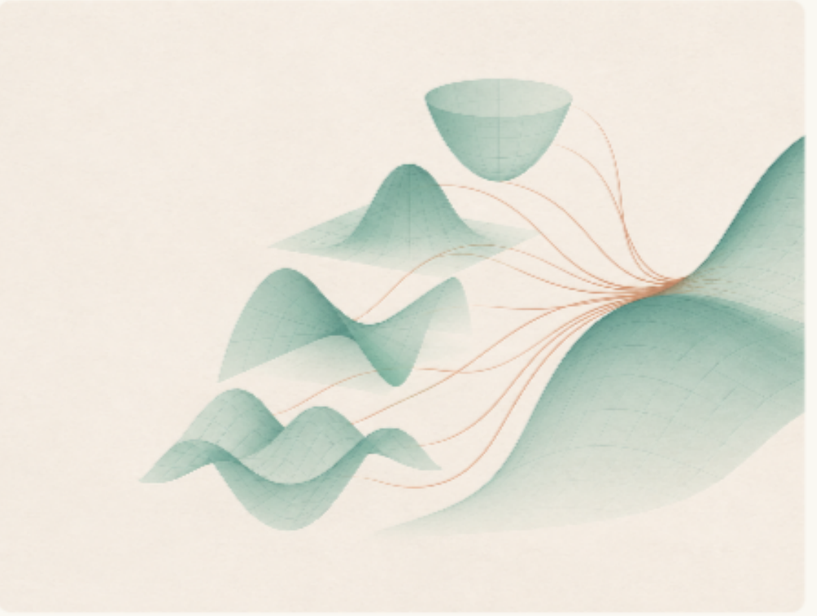
How machines learn

A delivery-time predictor

Saad Anjum, CTO
11 November 2026



An order arrives

The customer is 2 km from the restaurant.

How long will delivery take?

Distance is an input supplied by the order.

Preparation takes time

Our first preparation-time guess

3 minutes

An adjustable estimate: a parameter

Travel takes time

Our first travel-time guess

2 min / km

For this 2 km order: **4 minutes**

Our first prediction

3 minutes preparation + 4 minutes travel

Estimated delivery time

7 minutes

The delivery finishes

Observed delivery time

11 minutes

The same 2 km order / synthetic teaching example

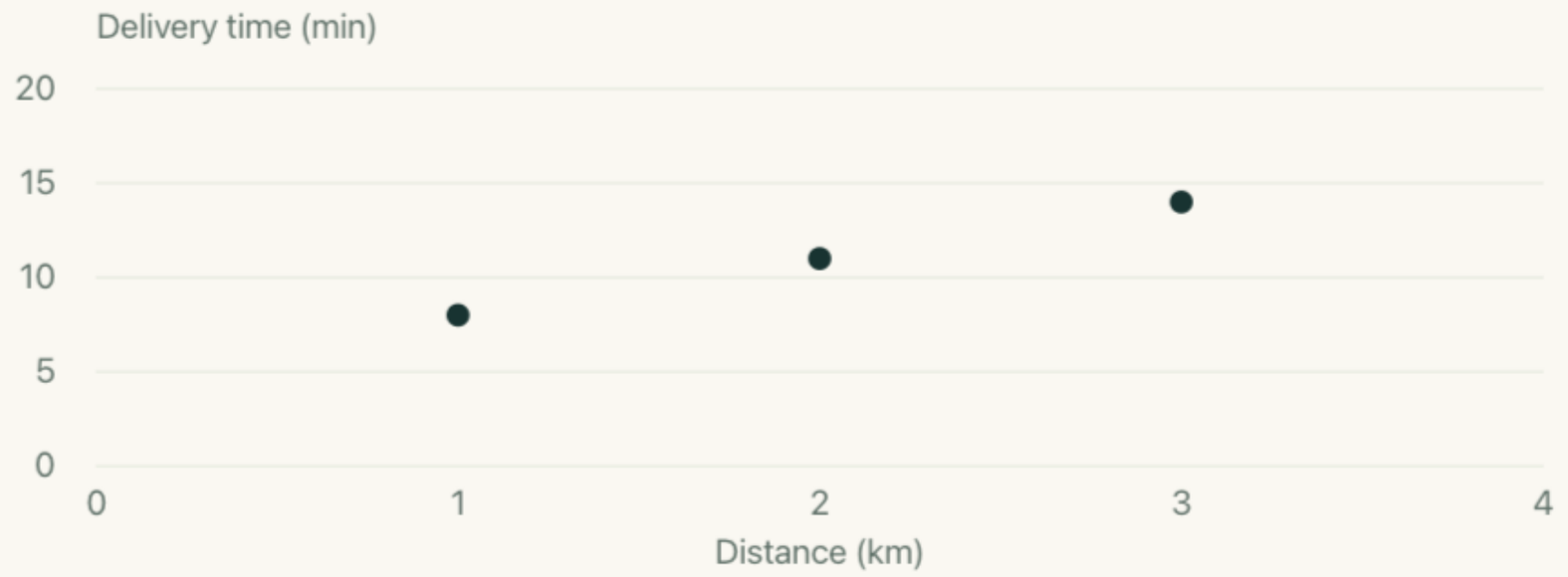
Feedback from several deliveries

Synthetic teaching records

Distance	Observed delivery time
1 km	8 minutes
2 km	11 minutes
3 km	14 minutes

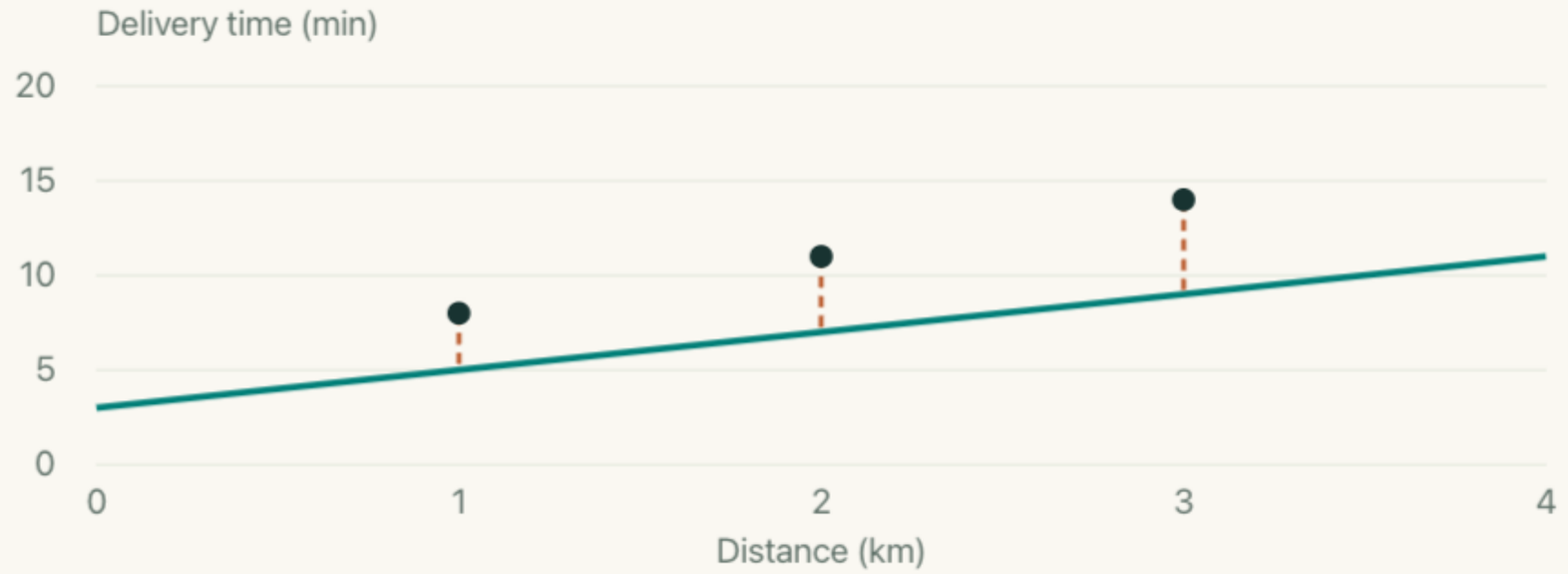
Our observations on a graph

Synthetic delivery observations



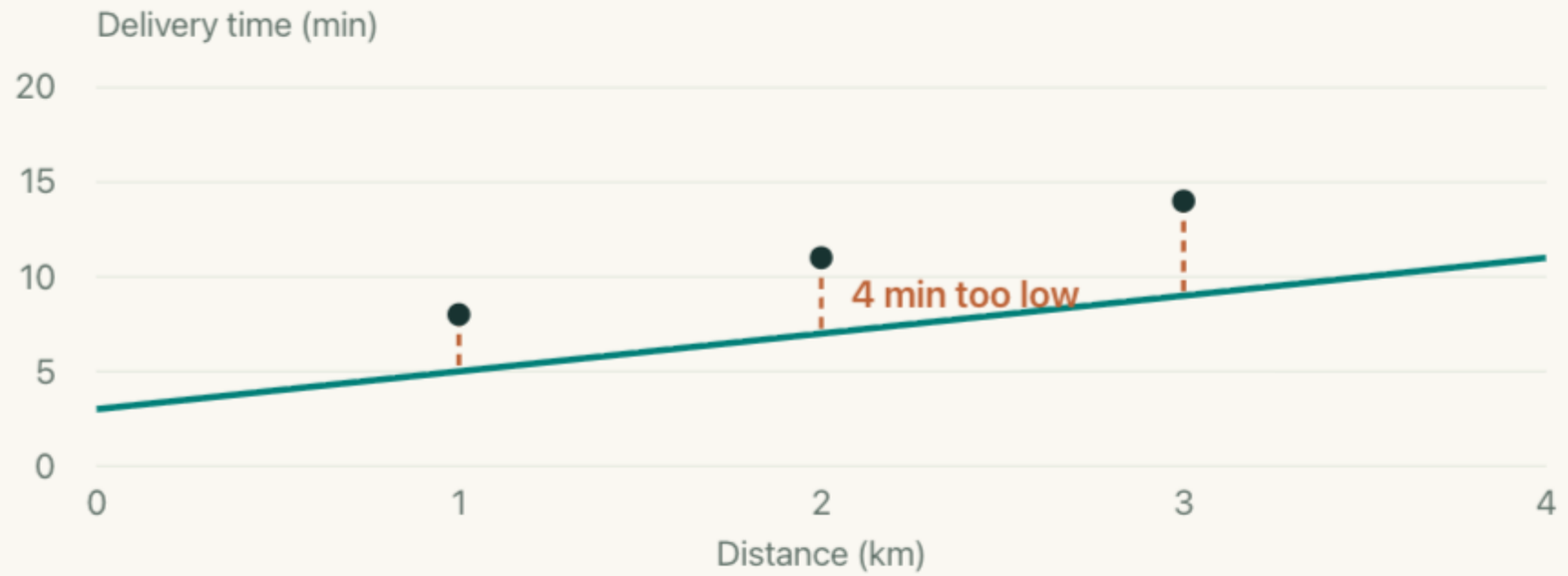
Our model makes a line

The same observations, with our first prediction rule




```
Estimated minutes = 3 + distance × 2
```

Measuring the mismatch



Each vertical gap is a **prediction error**.

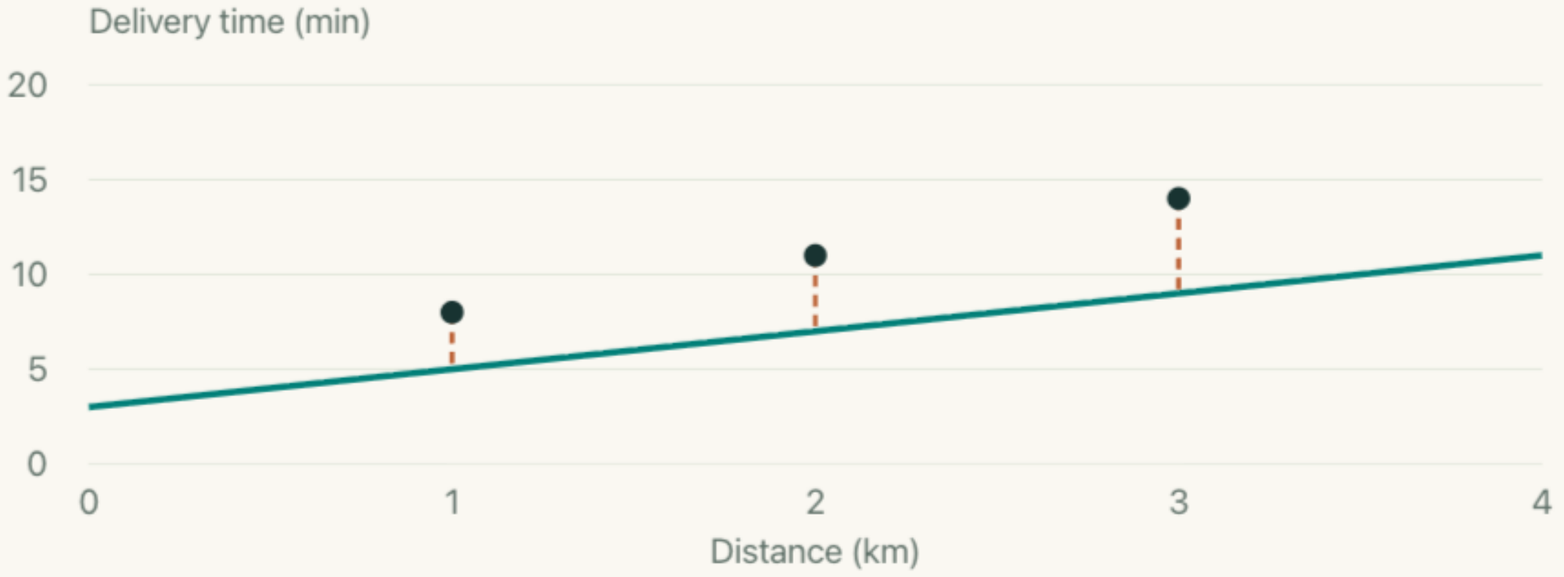
Loss is a score calculated from those errors.

Smaller is better.

Tuning our two parameters

`Estimated min = 3.0 + distance × 2.0`

Loss (MSE, min²) **16.6667**



Preparation / bias (minutes) 3

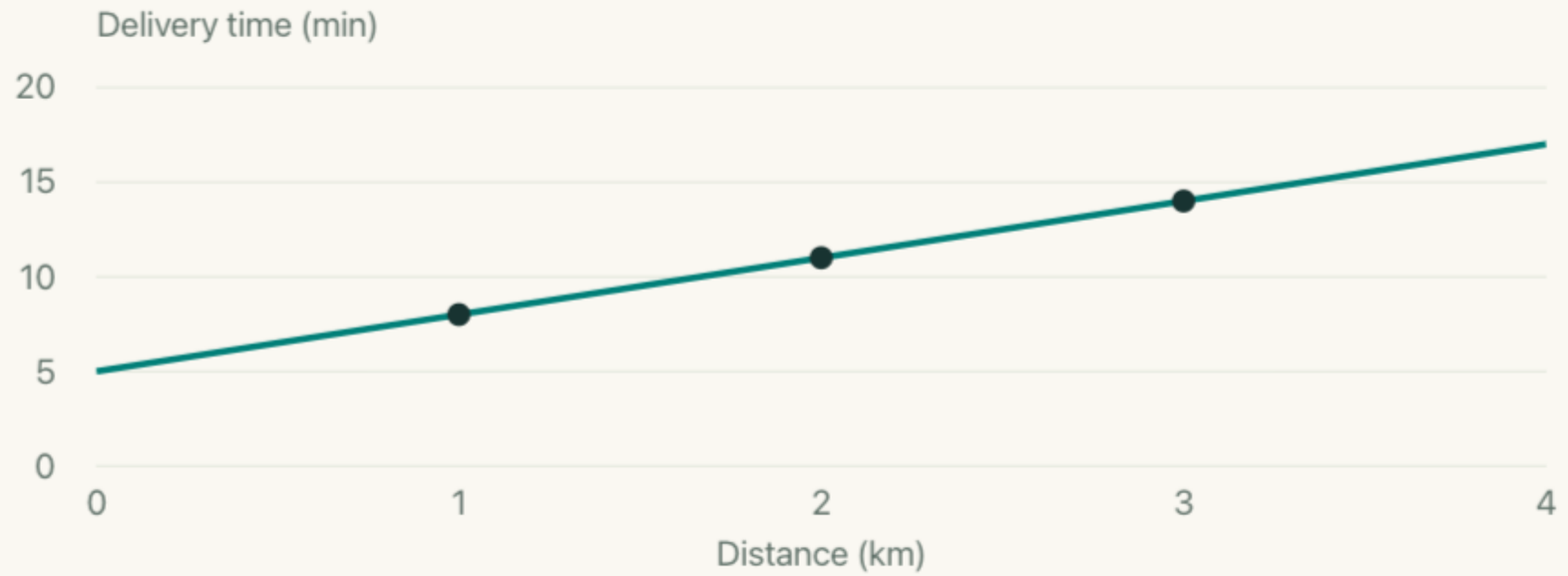
Travel rate / slope (min/km) 2

Loss is calculated across the same three observed deliveries.

Reset parameters

11

A better fit




```
Estimated minutes = 5 + distance × 3
```

Predictions: 8, 11, 14 minutes

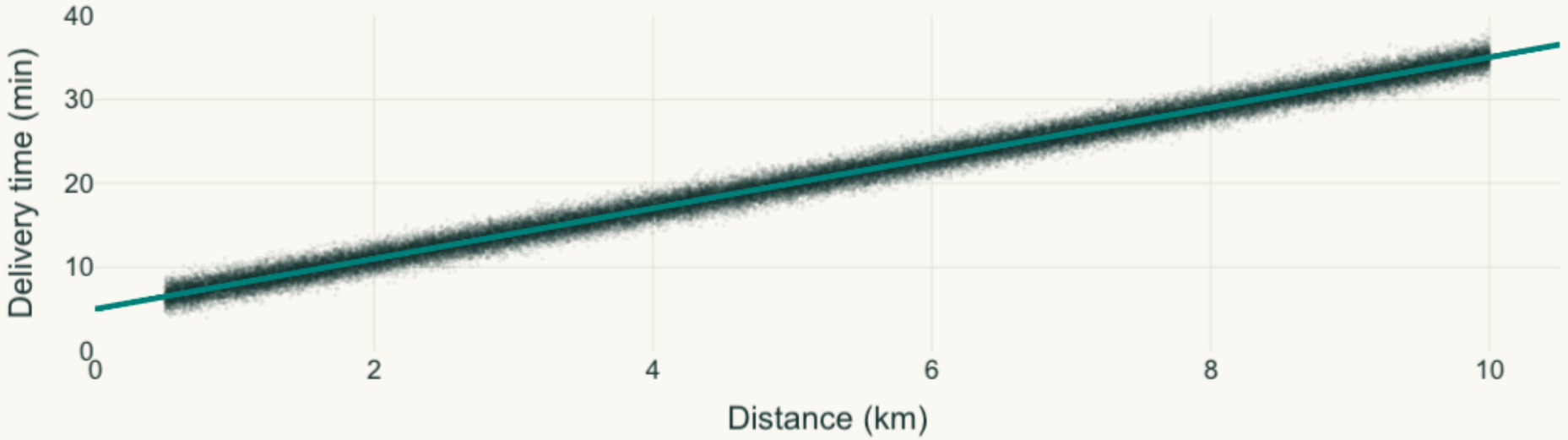
Who chooses the adjustments?

How could the program choose better parameter values?

Learning from a stream of deliveries

100,000 deliveries

Loss on fixed check sample: **0.822**



Synthetic observations / current prediction in teal

Complete

Run / Pause | Reset simulation

Starts again from our original guesses: 3 min and 2 min/km

Predicting numbers or categories

Linear regression

A straight-line model that predicts a number

Example: delivery time

11 minutes

Classification

Predict a category

Example: delivery status

On time or late